# Project 1 Report

Pin-Jing, Li. r1124995

## Task 1.

**1.a** We want to work in the quotient group $G \cong \mathbb{F}_{p^2}^{\times}/\mathbb{F}_p^{\times}$. Therefore, we construct the computation in G through the ring

$$R_N := (\mathbb{Z}/N\mathbb{Z})[x]/(x^2 + a), \quad R_N \bmod p \cong \mathbb{F}_{p^2}$$

if $a$ gives an irreducible polynomial. In $\mathbb{F}_p^{\times}$, there are half of the elements resulting in the desired $a$. The probability of it being irreducible, assuming uniformly sampling a in $\mathbb{F}_p$,

$$\frac{(p-1)/2}{p} = \frac{1}{2} - \frac{1}{2p}.$$

On average every two trials will give us a valid irreducible polynomial.

An element $v \in \mathbb{F}_{p^2}^{\times}$ correspond to a neutral element if $v \in \mathbb{F}_p^{\times}$, because the neutral element of the quotient is the subgroup being quotiented out.

$$v := a_0 + a_1 w. \quad v \in \mathbb{F}_p \Leftrightarrow a_1 = 0.$$

So we take $B!$-th power of a randomly sampled $u \in R_N$, let $v := u^{B!}$, and check $gcd(a_1, N) = 0$ to test the neutrality of $v$.

**1.b.** We similarly consider, since $p^3 - 1 = (p-1)(p^2 + p + 1)$, we work with $G \cong \mathbb{F}_{p^3}^{\times}/\mathbb{F}_p^{\times}$ and

$$R_N := (\mathbb{Z}/N\mathbb{Z})[x]/(x^3 + ax + b), \quad R_N \bmod p \cong \mathbb{F}_{p^3},$$

if the sampled $x^3 + ax + b$ just so happened to be irreducible. The probability of a monic cubic being irreducible is

$$\frac{p \cdot p^2 - \binom{p}{2} + \binom{p}{3}}{p^3} = \frac{1}{3} - \frac{1}{3p^2}.$$

Therefore, we are expected to take three trials. The neutral elements can be recognized similarly with $v := a_0 + a_1 w + a_2 w^2$. $v \in \mathbb{F}_p \Leftrightarrow a_1 = 0 \wedge a_2 = 0$. Therefore, we check $\gcd(a_1, N)$ and $\gcd(a_2, N)$ for neutrality.

**ECM and Higher Degree**.

For ECM, we randomly choose new elliptic curves and hope it gives a smooth $\#G$. With our construction, **[AI-generated, ChatGPT 5.2]** the group order is restricted to the form $p^2 + p + 1$. This gives us a structural group to work with, but sacrifices the flexibility available in ECM.

As the degree $n \to \infty$, the probability of sampling a irreducible polynomial is roughly $1/n$, making it less attractive.

**Better exponent choice than $B!$** Since we only need $B$, there exists a lot of redundancy if we directly use $B!$. Taking only primes $q \leq B$

$$M := \prod_{q \leq B} q^{\lfloor \log_q B \rfloor}$$

would be suffice. The efficiency improvement is made since we are taking a smaller exponent. Using Stirling's approximation on $B!$ and compare the bit-length,

$$\log(B!) \simeq B \log(B), \quad \log(M) \simeq \frac{B}{\log B} \cdot \log B,$$

it gives us roughly $\log(B)$ improvement when exponentiating.

**Listing Results.**

(a) $B = 5 \times 10^6$.

526645390122669789462387195963296956483700357011882308972655200768539072611077583239782963572216338498096466345412053818913899704980576764377382022974939892713651725406839879202190596766705196104829870800000296287022812770342507880227938889494768300380301333604981470837267502445569622501987608845550754759749 3

(b) $B = 5 \times 10^6$.

49731452329466260251982302847544607490963917 1

# Task 2.

The fixed padding scheme can be written as an affine equation. Let $u := k \| H(k)$, then

$$\mathrm{PAD}(k) = \sum_{i=0}^{t-1} 2^{256i} u + (2^{256} - 1) 2^{256t} =: Ru + C,$$

we therefore solve it with univariate Coppersmith for a small root. For a degree-$e$ polynomial, the small root condition is roughly

$$|u| = 2^{256} \leq N^{1/e},$$

so we require $e \leq \log_2 N/256 \simeq 4$ for this attack to be successed. The key we found for $e = 3$ is $k = 260475610619077747997499485920008829211$.

I wasn't able to finish the implementation, therefore only the idea was described in here. To handle larger values of $e$, we can treat this as a bivariate equation. Let $u = k + 2^{128}h$,

$$f(k, h) = (Ak + Bh + C)^e - c \equiv 0 \pmod{N}.$$

Heuristically this makes the solvable bound roughly

$$k \leq N^{1/2e}, h \leq N^{1/2e}.$$

Making those $e \leq \log_2 N/128 \simeq 8$ roughly lie in the solvable region. This formulation also made the PADR scheme solvable.

**Suggestion on Determinism and Modifying Padding Scheme.** The determinism structure is not the main issue for the desinged attack since 128 bit too large to be exhausively searched. It would be fatal if the message space is small.

This attack is available since the padding scheme results in an affine function of a small variable $u$. We can change the hash padding $\mathrm{PAD}(k) = k\|H(k\|1)\|k\|H(k\|2)\| \ldots \|111 \ldots 1$, making it either a multivariate coppersmith or a univariate coppersmith with long unknown part. **[AI-generated, ChatGPT 5.2]** This increases the effective entropy of the unknown and prevents direct application of small-root techniques.

# Task 3.

We are given a tuple $(N, a, r)$ such that $a^r \equiv 1 \pmod{N}$, where $r = \mathrm{ord}_N(a)$ is odd. Let $N = \prod_i p_i$ (possibly repeated primes) and $r_i = \mathrm{ord}_{p_i}(a)$. Then we have

$$r = \mathrm{lcm}(r_1, \ldots, r_i, \ldots),$$

and any divisor $d \mid r$, since for those $r_i \mid d \to a^d \equiv 1 \pmod{p_i}$,

$$\gcd(a^d - 1, N) = \prod_{i : r_i \mid d} p_i.$$

To construct our algorithm, we search for a divisor $d \mid r$ yielding nontrivial gcd. To avoid direct factoring on $r$ we enumerate through small primes $q$ and consider the divisor $d = r/q$, when $\gcd(a^d - 1, N) \neq\notin \{1, N\}$ we have a split on $N = g \cdot h$ where

$$g = \gcd(a^d - 1,\ N), \quad h = \frac{N}{g}.$$

For each prime factor of $g$ we therefore know $\mathrm{ord}_p \mid r/q$. We then recursively apply OrderToFactors to further split the two factors $(g,\ a \bmod g,\ r/q)$ and $(h,\ a \bmod h,\ r)$ until hitting a prime.

**Failure Analysis.** If any two primes have the exact same order, we would not be able to distinguish them with our extractions.

**Listing Results.** However, since our formulation essentially transform the factoring $N$ problem into a factoring $r$ problem, if $r$ is still a composite of large primes our algorithm would not be able to tackle. Typically in my attempts, I've tried enumerate up to $p \leq 10^9$ but did not managed to solve the latter two, suggesting that the factors of the two cases are beyond the range. The result for the first two are listed below.

case(a)

16219713666811825216880660196769518240719404774698650168925497830948941534933065844136295950553046136629685910973417819575225366642642655851385102382609141

98600611701057078723246096876363943023383776819687178966457025992673563864643957361337879230276482444881299572646145753359141186807903812940420954562207 61

case(b)

23607715957043363159568002864656809707664339181274142896378999942725382627074301909621717105537292103260226048176359097128073137598165893822914611869261639,

21528184269826598990271285307463507476524429426198897202841849683980585300943081595414831671675540572224596901940160205923379938264166169546053906691829687,

23390946821276445154821831004849720548838569753544934037279943353356600490015885001237840792849096195083107744858259649682892361948170384317217727144579591,

16301269261912676521092286863017811358695576078393559330072748652693628164666342314927897143502887501093311895612465091332761511251958813863269351674224 37

# Task 4.

**Decryption.** My design exploit that the difference between the closest vector of the lattice $\mathcal{L}(M)$ to our received codeword is the message.

$$M := \begin{bmatrix} \text{coeff}(f) & \text{coeff}(xf) & \dots & \text{coeff}(x^{n-1}f) & \text{coeff}(g) & \dots & \text{coeff}(x^{n-1}g) \end{bmatrix}^T \in \mathbb{Z}^{2n \times 2n}.$$

Thus, at the Rx side, we construct the lattice accordingly and run $a = \text{CVP}(\mathcal{L}(M), c)$, $\hat{m} = c - a$. Correct recovery requires the noise $m$ being a small noise. We therefore want $\|m\| \leq \frac{\lambda_1(\mathcal{L}(M))}{2}$. To ensure $\lambda_1$ not being too small, we want $B$ to not be too small.

**Plain Text Attack.** Similarly, by collecting the vectors from $R \geq 2n$ pairs $c_i - m_i$, we can reconstruct a basis that forms the same lattice, $\mathcal{L}_{\text{rec}} = \langle \text{coeff}(c_i - m_i) \rangle_{\mathbb{Z}} \subseteq \mathbb{Z}^{2n}$. We try to find vectors in $\mathcal{L}_{\text{rec}}$ that satisfy our key construction criterion. To extract out the low degree elements, we consider the low-degree part of the reconstructed lattice $\mathcal{L}_{\leq n}$. To compute this, we consider the partition of $\hat{M}$ a basis matrix of $\mathcal{L}_{\text{rec}}$, $\hat{M} = [H \mid T]$, where $H \in \mathbb{Z}^{r \times (n+1)}$ contains coefficients of degrees $0, \dots, n$, and $T \in \mathbb{Z}^{r \times (n-1)}$ contains coefficients of degrees $n+1, \dots, 2n-1$. A vector $v = z\hat{M}$ satisfies the degree restriction of a key if and only if its high-degree part vanishes, giving $zT = 0$.

Let $N$ be a basis matrix of the integer left nullspace of $T$. Then the matrix $M_{\text{low}} = N\hat{M}$ generates $\mathcal{L}_{\leq n}$. We then find short bounded vectors $v \in \mathcal{L}_{\leq n}$, and check whether it satisfy $\deg v = n$, $|v_i| \leq B$.

**Complexity.** LLL runs in $O(n^6 \log^3(UB))$ in this algorithm.

**Secret Key augmentation.** The computation load mostly lies in $CVP$ for decryption. To make CVP faster, we want to work with a LLL reduced $M$. More precisely, we want $M$ to be close to orthogonal. We might be able to borrow the idea from the cyclic constructions of BCH codes for this, to impose low correlation across the integer coefficients.

**Failure Analysis.** Our algorithms fail when the $R$ received pair generates a matrix with rank less than $2n$, or when there exists multiple bounded vectors in our final search.

**Cipher Text Only Attack.** Without knowing the plain texts, our problem becomes a lattice learning with bounded error problem, resembling an LWE with bounded noise being in the range of all possible messages.

# References

[1] Michael Artin. *Algebra*. Pearson, Boston, 2 edition, 2011.

[2] Steven D. Galbraith. *Mathematics of Public Key Cryptography*, volume 151 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2012.